METRIC

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26 May 1992
SUPERSEDING
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26 February 1992

MILITARY SPECIFICATION

HEATER ASSEMBLY, COMBUSTION VEHICULAR COMPARTMENT, 8.8/17.6 kW (30,000/60,000 BTU/HR), GENERAL SPECIFICATION FOR

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a forced air inlet personnel heater assembly, referred to herein as "heater" (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2540

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1.2 Classification. Heaters shall be of the following types and classes, as specified:

Type I	-	Heaters that must take their combustion and ventilating air from the same source ("Single Air").
Class I	-	Heaters that have a nominal rating of 8.8 kilowatts (kW) (30,000 British Thermal Units per Hour [Btu/hr]), but are also capable of operating at a nominal heating rate of 4.4 kW (15,000 Btu/hr).
Class II	-	Heaters that have a nominal maximum rating of 17.6 kW (60,000 Btu/hr), but are also capable of operating at a nominal heating rate of 8.8 kW (30,000 Btu/hr).
Type II	-	Heaters that can take their combustion and ventilating air from separate sources ("Dual Air").
Class II	-	Heaters that have a nominal maximum rating of 17.6 kW (60,000 Btu/hr), but are also capable of operating at a nominal heating rate of 8.8 kW (30,000 Btu/hr).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-T-266	- Thinner: Dope and Lacquer (Cellulose-Nitrate).
VV-F-800	- Fuel Oil, Diesel.

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MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-T-13513	- Terminal, Lug; Splice, Conductor (Electrical, Solderless for Automotive use).
MIL-B-43436	- Band, Marker Blank.
MIL-F-45356	- Filter, Fluid, Pressure, and Filter Element, Fluid, Pressure: for Gasoline, Internal Combustion Engines, with Carburetor.
MIL-H-62550/1	- Heater Assembly, Combustion, Vehicular Compartment, 17.6 kW, Single Air Type (Metric).
MIL-H-62550/2	- Heater Assembly, Combustion, Vehicular Compartment, 17.6 kW, Dual Air Type (Metric).
MIL-H-62550/3	- Heater Assembly, Combustion, Vehicular Compartment, 8.8 kW, Single Air Type (Metric).
MIL-C-81562	- Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited).
MIL-M-81594	- Marking Foil, Hot Stamp Printing of Electrical Insulating Materials.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Constuction Equipment, Material Handling Equipment, and Spare Parts
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-1261	- Arc Welding Procedures for Construction Steel
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-45662	- Calibration System Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

11669705	- Box Assy, Control.
12268109	- Box Assembly, Accessory Control.
12330264	- Control Box, Electrical, Heat-Ventilator.
12345539	- Driver's Master Panel Assembly.

PURCHASE DESCRIPTIONS

ATPD 2165	- Heater Test Stand, Valcom Model VHTS-89102

(Copies of specifications, standards, drawings, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of

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the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Performance Test Code - Application Part II of Fluid Meters Sixth (PTC) 19.5 Edition 1971 (Interim Supplement 19.5 on Instruments and Apparatus).

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 E. 47th Street, New York, NY 10017).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.
ASTM D439	- Automotive Gasoline.
ASTM D2156	- Smoke Density in Flue Gases from Burning Distillate Fuels, Test for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

210-74	- Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to the Air Movement and Control Association, 30 W. University Drive, Arlington Heights, IL 60004).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

SAE J726	- Air Cleaner Test Code.

(Applications for copies should be addresses to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for related associated detail specifications, specification sheets, or MS sheets), the text of this document

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takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3).

3.2 Scope. The heater shall meet the physical (see 3.3), environmental (see 3.4), performance (see 3.5), and reliability/durability (see 3.6), requirements as specified herein. The heater shall be constructed to the form and dimension depicted on the applicable specification sheet (see 6.2) and shall be compatible with control box 11669705, 12268109, 12330264, panel assembly 12345539 and test stand ATPD 2165 (see 4.7.1).

3.3 Physical.

3.3.1 Weight. Class I heaters shall not weigh more than 9.5 kilograms (kg) (21 pounds [lbs]). Class II heaters shall not weigh more than 17 kg (38 lbs) (see 4.7.3).

3.3.2 Material.

3.3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, the compatibility of dissimilar metals shall be in accordance with MIL-STD-889 (see 4.7.1).

3.3.2.2 Steel components. Nonpainted steel components, other than stainless steel, shall be zinc plated in accordance with ASTM B633, type II, class FE/Zn 13 or MIL-C-81562, type II, class 5. Heat radiating surfaces may be aluminized steel (see 4.7.1).

3.3.2.3 Aluminum components. Aluminum components shall be coated in accordance with MIL-C-5541, class 1A (see 4.7.1).

3.3.2.4 Radioactive material. Radioactive material shall not be incorporated into the heater assembly (see 4.7.1). Radioactive material for the purpose of this specification, is any material with a specific activity greater than 0.002 microcurie per gram or with an activity greater than 0.01 microcurie for any component.

3.3.2.5 Electrical components and wiring. The wire and insulation shall be flexible to -54 degrees Celsius (°C) [-65 degrees Fahrenheit (°F)]. Each wire end or connector shall be uniquely marked in accordance with MIL-B-43436 or MIL-M-81594. Marking the igniter leads is optional if they are readily identifiable. Crimped terminals shall be crimped according to

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MIL-T-13513. Soldering shall conform to MIL-STD-454, requirement 5 (see 4.7.1).

3.3.2.6 Filters. A filter shall be placed in the fuel line with a sieve rating of not less than 210 microns and shall have a minimum surface area of 0.25 square centimeters (cm^2) [0.04 square inches (in^2)] (see 4.7.1).

3.3.3 Interchangeability. MIL-H-62550/2 heaters shall be easily convertible to MIL-H-62550/1 through the use of screw drivers, hex keys or hex sockets. The conversion of heaters shall not remove any protective or sensing devices needed to fulfill requirements specified herein. After conversion, dimensional and functional conformance to MIL-H-62550/1 shall be verified (see 4.7.1 and 4.7.4).

3.3.4 Welding. All heater parts that are exposed to combustion gases during normal operation shall be welded, as applicable, according to MIL-STD-1261, Class II (see 4.7.1).

3.3.5 Combustion air inlet strength. Type II only. The heater's combustion air inlet shall be capable of sustaining at least 50 kg (110 lbs) load without permanent bending, cracking, interference between moving parts, or any damage that would effect heater performance or durability. Combustion air inlet adapter rotating torque increase shall be less than 20% at the end of the test (see 4.7.5).

3.3.6 Safety. Surface areas of the heater outside of designated hot spots shall not cause contact burns (see 4.7.1).

3.3.7 Protective coating. The heater shall be painted in accordance with MIL-STD-193 (see 4.7.1).

3.3.8 Identification and marking. Nameplates in accordance with MIL-P-514, type III, composition A, class 2 shall be permanently attached to the heater with metal fasteners. Nameplates shall be marked in accordance with MIL-STD-130 with the following information (see 4.7.1 and 4.7.2):

a. Design activity CAGE and Part or Identifying Number (PIN).
b. Manufacturer's name, CAGE, and model number.
c. Contract number.
d. Nomenclature: Heater, Vehicle Compartment, Fuel-Fired, 17.6 kW (60,000 Btu/hr) or 8.8 kW (30,000 Btu/hr), as applicable.
e. National stock number.
f. Serial number.
g. Type II (only) heaters. Combustion air inlet adapter shall be marked as follows with letters at least 1.25 cm (0.5 in) high by black paint: "DO NOT REMOVE".

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3.4 Environmental conditions. The heater shall meet the requirements of 3.5 after being subjected to the environmental conditions specified in 3.4.1 through 3.4.8.

3.4.1 Temperature.

3.4.1.1 High temperature. Storage at a temperature of at least 71°C (160°F) and operation with combustion air temperature of at least 50°C (120°F) (see 4.7.6.1).

3.4.1.2 Low temperature. Storage at -54°C (-65°F) and operation down to -50°C (-55°F) (see 4.7.6.2).

3.4.2 Humidity. Exposure to temperature and humidity cycles specified by MIL-STD-202, method 106 (see 4.7.7).

3.4.3 Salt fog. Exposure to salt fog for 96 hours (see 4.7.8).

3.4.4 Shock. Exposure to 40 gravity units (g) shock loading for 18 milliseconds (see 4.7.9).

3.4.5 Vibration. Exposure to sinusoidal vibration in accordance with figure 1, including up to four resonance dwells (see 4.7.10).

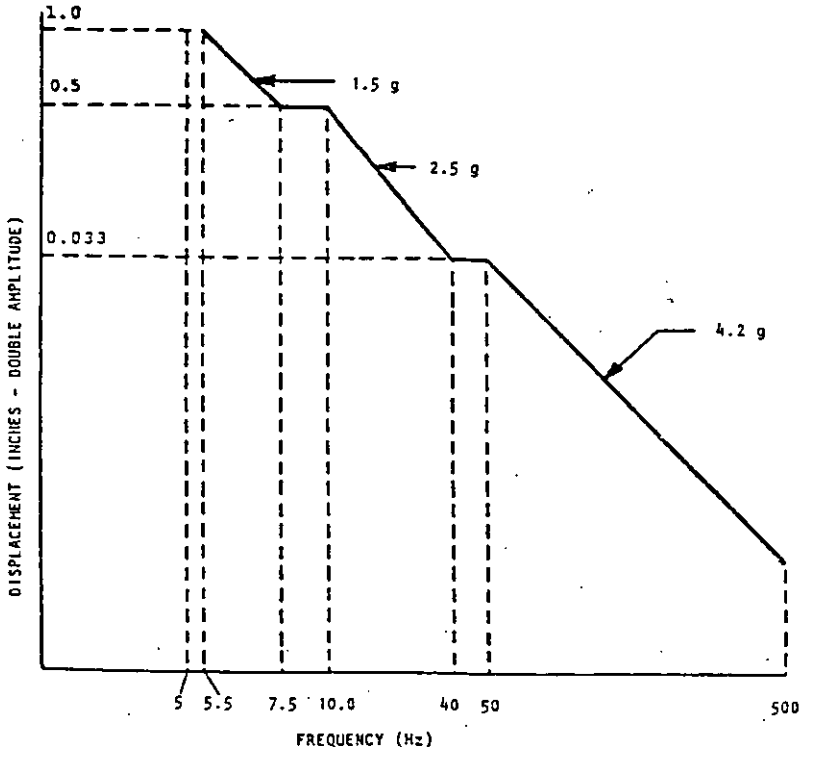


FIGURE 1. Vibration curve.

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3.4.6 Electromagnetic interference (EMI). The heater shall meet the requirements of MIL-STD-461A, notice 4 (CE-04 and RE-02) (narrow and broad band), and operate as specified (see 3.5.7 and 3.5.1.1) when exposed to electrical interference specified by CS-01, RS-03 and RS-03.1 for all operating modes including the start cycle. The single switching actions that initiate and end the heater start cycle, do not have to meet CE-04 and RE-02 EMI requirements (see 4.7.11).

3.4.7 Dust. Exposure for at least 200 hours with at least 3 grams per cubic meter (g/m^3) of fine dust per SAE J726 in the combustion air and at least 300 grams coarse dust per SAE J726 in the fuel (see 4.7.12).

3.4.8 Washdown. Class II only. Exposure to simulated vehicle washdown test as specified in 4.7.13. During the start cycle, if the heater fails to start, the control box run indicator shall flash and the heater shall enter its self diagnostic mode as specified in 3.5.2.

3.5 Performance. The heater shall meet the performance requirements specified herein when mounted in any position from horizontal to vertical with the air inlet level with, or higher than, the air outlet; and, except as otherwise specified, when operating in the high, and low heat modes.

3.5.1 Control circuits.

3.5.1.1 Start cycle. The heater control circuits shall control the heater start cycle and, upon starting, automatically switch to run mode and illuminate the control box run indicator. The starting time, measured from the time the control box "run-start" switch is put in the "start" position until the igniter circuit is de-energized shall not exceed 3.5 minutes (see 4.7.14).

3.5.1.2 Starting with fault. If the heater fails to start in the allotted time (see 3.5.1.1), the heater shall automatically exit the start mode, diagnose the fault, and exhaust any fuel vapors (see 3.5.13). The heater shall not be damaged when subjected to the conditions described in a, b, and c below. For conditions b and c only, there shall be less than 140 cubic centimeters (cc) (class I) or 240 cc (class II) of total fuel, it shall enter its purge cycle, shut off the fuel flow, and then enter its self diagnostic mode in accordance with 3.5.2 (see 4.7.15).

a. Heater control box start switch in the "start" position for more than 30 minutes.
b. Open igniter circuit.
c. Shorted igniter circuit.

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3.5.2 Self diagnostics. The heater shall indicate support system and repair part faults by flashing the control box run indicator. Individual faults shall be identified by the heater, at the heater, without the need for reference to additional documentation by maintenance personnel (see 4.7.16).

3.5.3 Fuel.

3.5.3.1 Fuel compatibility. The heater shall be capable of operating on JP-5 per MIL-T-5624, JP-8 per MIL-T-83133, and DF-1, DF-2 and DF-A per VV-F-800 (see 4.7.17).

3.5.3.2 Fuel consumption. Fuel consumption shall not exceed the amounts shown by table I (see 4.7.18).

TABLE I. Maximum fuel flow.

Control box heat level setting	g/min 1/ (lbs/min) 2/ Class I		g/min Class II	(lbs/min)
High heat	22.2	(0.049)	41.2	(0.092)
Low heat	14.1	(0.031)	23.6	(0.052)

1/ g/min = grams per minute. 2/ lbs/min = pounds per minute.

3.5.3.3 Fuel leakage. There shall be no leakage from the heater fuel components (see 4.7.19).

3.5.3.4 Low fuel pressure. When fuel pressure drops below 28 kilopascals (kPa) (4.0 lbs per in^2 gage [psig]) the control box run indicator shall flash and the heater shall enter the self diagnostics mode according to 3.5.2 (see 4.7.20).

3.5.4 Voltage. The heater shall start and meet performance requirements with applied voltages of 23 to 30 volts (V) direct current (dc). At ambient temperatures below -32°C (-25°F) the heater shall start and run with an applied voltage of 19 Vdc (see 4.7.21).

3.5.4.1 Transient voltage. The heater shall not be damaged when exposed to electrical spikes and surges described by MIL-STD-1275 for both fault free and single fault operation of vehicle electrical system (see 4.7.22).

3.5.5 Current. At 24 Vdc, the current draw from the power source shall not exceed that specified by table II (see 4.7.23).

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TABLE II. Maximum current draw.

Ambient Temperature °C (°F)	Class I Start (A) 1/	Class I Run (A)	Class II Start (A)	Class II Run (A)
Above 7.2 (45)	12.0	8.0	16.0	15.0
Below 7.2 (45)	17.0	13.0	21.0	20.0

1/ A = Amperes.

3.5.6 Power loss. When power loss occurs there shall be less than 140 cc (class I) or 240 cc (class II) of total fuel flow. The heater shall not enter the start cycle or operate until the control box's "start-run" switch is placed in the "start" position (see 4.7.24).

3.5.7 Heat output. The heat output of the heater shall be as shown by table III. High and low heat shall be manually setable by the control box (see 4.7.25).

TABLE III. Heat output.

Control box heat level setting	Class I kW 1/ (Btu/hr) 2/	Class II kW (Btu/hr)
High heat	8.8 + 0.9 - 0.9 (30,000 + 3,000 - 3,000)	17.6 + 3.6 - 1.8 (60,000 + 12,000 - 6,000)
Low heat	4.4 + 0.9 - 0.9 (15,000 + 3,000 - 3,000)	8.8 + 1.8 - 0.9 (30,000 + 6,000 - 3,000)

1/ kW = kilowatt 2/ Btu/hr = British thermal units per hour

3.5.8 Ventilating air discharge rate. With 24 Vdc applied, the ventilation air discharge rate at the restrictions specified in table IX or free flow shall be as specified by table IV. Test results shall be corrected to standard air (see 4.7.26):

Density = 1.20 kg per cubic meter (kg/m^3) (0.075 lbs per cubic foot [lbs/ft^3])
Pressure = 101.3 kPa (29.92 inches of Mercury (in Hg))
Temperature = 20°C (68°F)
Relative Humidity = 50%

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TABLE IV. Ventilation and combustion air discharge rates.

Minimum ventilation air flow		Specified air circuit restrictions			
		Class I		Class II	
kl/min1/	(scfm)2/	Ventilation	Combustion	Ventilation	Combustion
3.26	(115)	free flow	free flow	N/A	N/A
2.27	(80)	table IX	table IX	N/A	N/A
(low heat)					
5.38	(190)	N/A	N/A	free flow	free flow
4.81	(170)	N/A	N/A	table IX	table IX
(high heat)					
6.37	(225)	N/A	N/A	free flow	free flow
5.81	(205)	N/A	N/A	table IX	table IX

1/ kl/min = kiloliter per minute.
2/ scfm = standard cubic feet per minute.

3.5.8.1 Ventilation. Class II heaters only. When the heater is used in the ventilation mode the ventilation circuit shall meet the air flow rate as specified in 3.5.8. The air flow rate in the combustion circuit shall not exceed 28 milliliters per minute (ml/min) (0.001 scfm) (see 4.7.27).

3.5.8.2 Seal leakage. When Class II heater is attached to a duct with a diameter between 202 and 208 millimeters (mm) (7.94 and 8.20 inches) and operated on high heat the ventilation air leakage shall be less than 85 liters per minute (l/min) (3 scfm) (see 4.7.28).

3.5.8.3 Ventilating air restriction. When the heater's ventilating air circuit is restricted, the heater shall limit the discharge air temperature between 150 and 177°C (300 and 350°F). When the air flow is restricted to less than 1.27 ± 0.42 kl/min (45 ± 15 scfm) (class I) or 2.55 ± 0.85 kl/min (90 ± 30 scfm) (class II) the ventilating air temperature shall not exceed 177°C (350°F), the heater shall shut off its fuel flow, enter its purge cycle, and then shut itself off (see 4.7.29).

3.5.9 Inlet over pressure. Heaters shall start, operate and have a combustion air flow of not more than 623 l/min (22 scfm) and meet the requirements of 3.5.1.2 when the inlet of the combustion air circuit is subjected to at least 2.0 kPa (8.0 in H_2O) pressure (see 4.7.30).

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3.5.10 Combustion.

3.5.10.1 Combustion air flow. Heater shall shut off its fuel flow and enter the self diagnostic mode when subjected to a reverse pressure of at least 1.5 kpa (6.0 in H_2O) or a reduced combustion air flow of less than 280 l/min (10 scfm) (see 4.7.31). The exhaust smoke density throughout shall not exceed a smoke spot number of three per ASTM D2156.

3.5.10.2 Combustion exhaust. Combustion exhaust temperature shall not drop below 260°C (500°F) (see 4.7.32).

3.5.10.3 Combustion vapor exposure. There shall be no combustion outside the heater when combustible vapors (winter grade gasoline, lacquer thinner or propane) are added to the heater's combustion air (see 4.7.33).

3.5.11 Over temperature power shut off. When the heater shuts off its fuel flow from excessive ventilation air temperature (see 3.5.8.3), voltage at pin C of the heater's connector shall drop to less than 2 Vdc (see 4.7.34).

3.5.12 Smoke density. After the heater has been operating for not longer than 7 minutes, its exhaust gases shall not exceed a smoke spot number of three per ASTM D2156 (see 4.7.35).

3.5.13 Purging cycle. The purging time shall not exceed 4.0 minutes and the exhaust temperature shall not be greater than 177°C (350°F) when the blower shuts off (see 4.7.36).

3.5.14 Toxic fumes. Heat exchangers shall meet the leak requirements of MIL-STD-1184, type II, class II components, except substitute procedure 5 for procedure 1. After completion of environment and durability testing the heat exchanger's leakage shall not exceed 14 l/min (0.5 scfm) (see 4.7.37).

3.6 Reliability and durability. During the reliability and durability verification 4.7.38 and testing 4.7.39, the heater shall operate on contaminated diesel fuel. During testing the heater shall meet the requirements of 3.5.1.2, 3.5.4, 3.5.5, 3.5.8, 3.5.10.2 and 3.5.14 and run with transient voltages down to 19 Vdc. After completion of each phase the heater shall meet the requirements of 3.5.3.2, 3.5.3.3, 3.5.4, 3.5.7, 3.5.8, 3.5.11 and 3.5.12.

3.6.1 Reliability. The heater shall have a minimum reliability of 1250 mean hours between failure (MHBF). A reliability failure is defined as any malfunction of the heater that requires component replacement above unit level.

3.6.2 Durability. The heater shall have a minimum durability requirement of 2500 operating hours with at least 2400 start

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cycles without failures. A durability failure is defined as any malfunction of the heater that is not repairable at direct support.

3.6.2.1 Reduced durability. For control test only, the durability testing shall be reduced to 300 cycles (see 4.7.39.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor may use their own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification.

4.1.1 Responsibility for compliance. Heaters shall meet all requirements of sections 3 and 5. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance shall comply with all requirements of the contract. The supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the parameter's tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

a. Air temperature: 23 ± 10°C (73 ± 18°F).
b. Barometric pressure: 96.5 ± 6.8 kPa (28.5 ± 2 in Hg).
c. Relative humidity: 10 to 90 percent.

4.3 First article inspection. The first article test plan shall be approved by the procuring activity prior to beginning of first article testing. The Government shall randomly select three of the first ten production heaters produced under the production contract for first article inspection. First article samples shall be numbered and inspected as specified in 4.3.1 and in table V. Approval of the first article sample shall not relieve the contractor of his obligation to supply heaters that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer. Unless otherwise specified, the listing in table V does not imply any particular test sequence. Failure of

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the first article to meet any of the requirements specified herein will be cause for the Government to refuse to perform surveillance on additional testing until the fault(s) have been identified and corrected to the Government's satisfaction (see 3.1).

4.3.1 In-process inspections. During fabrication of the first sample production heater, an in-process inspection shall be conducted by the contractor under the surveillance of the Government inspector. Inspection shall be made prior to application of primer and paint. Processing and welding procedures, quality system, and inspection records shall be reviewed and evaluated during this inspection.

4.3.2 Production sample. After completion of inspection, the first sample production heater shall remain at the manufacturing facility, as a production sample, and be the last heater shipped on the contract. The heater subject to durability testing shall not be used to fulfill production quantities of the contract.

4.3.3 First article inspection failure. The procuring agency shall determine whether a particular first article failure shall be treated as major or minor.

a. A major failure is a non-compliance with requirements specified herein by a functioning component, heat exchanger, or when corrective action requires a design change. As a minimum, the durability or environmental test during which the deficiency surfaced shall be repeated. A complete first article test repeat of any or all units may be necessary at the discretion of the procuring agency.

b. A minor failure, in general, involves a non-compliance caused by a non-functioning component, a minor workmanship deficiency, or when design change is unnecessary. At a minimum, a repeat of the durability or environmental test which caused the failure shall be required.

c. The Government shall not accept delivery of any production units until all first article requirements have been satisfied.

4.4 Quality conformance inspections (QCI). QCI shall consist of the examinations of 4.4.2 and the acceptance tests of 4.4.3. Nonconformance with any of the specified requirements in sections 3 and 5 shall be cause for rejection.

4.4.1 Lot formation. An inspection lot shall consist of all the heaters of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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TABLE V. Classification of inspections.

Title	Requirement	Inspection	First article test 1 Hor	2 Vert	3 Hor	Quality conformance Acceptance 100% Horizontal	Control Hor
Materials, design and construction	3.2, 3.3.2, 3.3.3, 3.3.4, 3.3.6, 3.3.8	4.7.1	X	X	X		X
Defects	3.2 3.3.7 3.3.8	4.7.2	X	X	X		X
Weight	3.3.1	4.7.3	X	X	X		X
Interchangeability	3.3.3	4.7.4	X				
Combustion air inlet strength	3.3.5	4.7.5	X	X	X		X
High temperature	3.4.1.1	4.7.6.1	X	X	X		X
Low temperature	3.4.1.2	4.7.6.2	X	X	X		X
Humidity	3.4.2	4.7.7		X			
Salt fog	3.4.3	4.7.8			X		
Shock	3.4.4	4.7.9	X		X		X
Vibration	3.4.5	4.7.10		X	X		X
Electromagnetic interference	3.4.6	4.7.11			X		
Dust	3.4.7	4.7.12	X		X		
Washdown	3.4.8	4.7.13	X		X		
Starting cycle	3.5.1.1	4.7.14	X	X	X	X	X
Starting with fault	3.5.1.2	4.7.15	X	X	X		X
Self diagnostics	3.5.2	4.7.16					
Fuel compatibility	3.5.3.1	4.7.17					
Fuel consumption	3.5.3.2	4.7.18	X	X	X	X	X
Fuel leakage	3.5.3.3	4.7.19	X	X	X	X	X
Low fuel pressure	3.5.3.4	4.7.20	X	X	X		X
Voltage	3.5.4	4.7.21	X	X	X	X	X
Transient voltage	3.5.4.1	4.7.22	X	X	X		X
Current	3.5.5	4.7.23	X	X	X	X	
Power loss	3.5.6	4.7.24			X		X
Heat output	3.5.7	4.7.25	X	X	X	X	X
Ventilating air discharge rate	3.5.8	4.7.26	X	X	X	X	X
Ventilation	3.5.8.1	4.7.27	X				
Seal leakage	3.5.8.2	4.7.28	X				X
Ventilation air restriction	3.5.8.3	4.7.29	X	X	X	X	X
Inlet over pressure	3.5.9	4.7.30	X				
Combustion air flow	3.5.10.1	4.7.31		X	X		
Combustion exhaust	3.5.10.2	4.7.32	X				
Combustion vapor exposure	3.5.10.3	4.7.33	X	X			X
Over temperature power shut off	3.5.11	4.7.34	X				X
Smoke density	3.5.12	4.7.35	X	X	X	X	X
Purging cycle	3.5.13	4.7.36	X	X	X	X	X
Toxic fumes	3.5.14	4.7.37	X	X	X	X	X
Reliability and durability	3.6	4.7.38 4.7.39	X	X			
Reduced durability	3.6.2.1	4.7.39.1					X

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4.4.1.1 Sampling for inspection. Sample size for quality conformance inspection shall be determined in accordance with table VI. Before sampling may be initiated, the contractor shall inspect 100% of the first 100 heaters to establish that the process average (see 6.8) falls within the control limits.

TABLE VI. Sampling inspection.

Sampling lot size	Sample size for Majors	Sample size for Minors
2 to 8	*	3
9 to 25	13	3
26 to 50	13	5
51 to 90	13	6
91 to 150	13	7
151 to 280	20	10
281 to 500	29	11
501 to 1200	34	15

* Indicates entire lot must be inspected.
NOTE: The acceptance number in all cases is zero.

4.4.2 Examination. Sample units shall be examined for the defects specified in table VII.

4.4.2.1 Failure disposition. Any defect shall be cause for the Government to refuse acceptance of the production represented by the sample size.

TABLE VII. Classification of defects.

Category	Defect	Method of examination
Major		
101	Assembly, incomplete (see 3.1).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.1).	SIE 1/
Minor		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.1).	SIE
202	Protective coating, improper (see 3.3.7).	Visual
203	Identification marking, improper (see 3.3.8).	Visual

1/ SIE = Standard Inspection Equipment.

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4.4.3 Acceptance test. Shall be conducted in accordance with the acceptance tests specified in table V (table VIII does not apply). Each heater shall be 100% inspected for each requirement. Any defect shall be cause for the Government to refuse acceptance until action taken by the contractor to correct defects and cause has been approved by the procuring agency. Acceptance testing shall be conducted at 23.0 ± 1.0 Vdc and high heat, control box optional.

4.5 Control tests. Control tests shall be conducted at a rate of one out of 600 heaters consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. Heaters shall be subjected to the control tests specified in table V. Failure of any heater to pass any of the specified tests shall be cause for the Government to refuse acceptance of the quantity represented until action is taken by the contractor to correct defects and to prevent recurrence has been approved by the procuring agency.

4.6 Inspection of packaging. The preservation, packaging, packing, and marking shall be inspected to determine conformance to 5.1.

4.7 Methods of inspection. Unless otherwise specified tests shall be conducted as specified in table V and in the sequence specified in table VIII, at the ambient conditions specified in 4.2 at the specified heat outputs, high and low heat, (see 3.5.7), and at input voltages (measured at the power input to the control box) of 23.0 ± 0.5 Vdc and 30.0 ± 0.5 Vdc. The fuel pressure shall be 55 ± 7 kPa (8 ± 1 psig). Temperature sensor shall be shielded from radiant heating by the heater. All tests shall be conducted with heater control box part number 11669705 with a nonshielded copper wire cable assembly 3 to 3.5 meters (10 to 12 ft) long connecting the control box with the heater. Wire gauge shall be equal or greater to that used in the control box for the same circuit. If the test set up requires a longer cable, the gauge of the wire and cable length shall be changed so as to provide equal resistance to that of the specified cable. No repairs, adjustments or maintenance may be performed upon the heater during the testing specified herein. When control box run indicator flashing is specified, the heater shall be monitored for identification of specific support system fault causing the flashing (see 3.5.2).

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TABLE VIII First article and control test sequence.

Inspection	Requirement	Title
4.7.37	3.5.14	Toxic fumes
4.7.6.1	3.4.1.1	High temperature
4.7.18 4.7.28	3.5.3.2 3.5.8.2	Fuel consumption Seal leakage
4.7.7 4.7.8 4.7.9 4.7.10 4.7.12 4.7.24 4.7.31.1	3.4.2 3.4.3 3.4.4 3.4.5 3.4.7 3.5.6 3.5.10.1	Humidity Salt fog Shock Vibration Dust Power loss Reverse flow
4.7.20 4.7.33	3.5.3.4 3.5.10.3	Low fuel pressure Combustion vapor exposure
4.7.6.2 4.7.13 4.7.14 4.7.19 4.7.21 4.7.23 4.7.25 4.7.26 4.7.29 4.7.36	3.4.1.2 3.4.8 3.5.1.1 3.5.3.3 3.5.4 3.5.5 3.5.7 3.5.8 3.5.8.3 3.5.13	Low temperature Washdown Starting cycle Fuel leakage Voltage Current Heat output Ventilating air discharge rate Ventilation air restriction Purging cycle
4.7.38 4.7.39	3.6	Durability and reliability
4.7.37	3.5.14	Toxic fumes

Note: Sequence of inspections within blocks is not mandatory.

4.7.1 Materials, design and construction. To determine conformance to 3.2, 3.3.2.1 through 3.3.4, 3.3.6 and 3.3.8, inspect contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.7.2 Defects. Conformance to 3.2, 3.3.7 and 3.3.8 shall be determined by examination for the defects listed in table VII.

4.7.3 Weight. To determine conformance to 3.3.1, weigh the heater.

4.7.4 Interchangeability. To determine conformance to 3.3.3 remove the dual air adapters using the specified tools. Subject the heater to the tests of 4.7.14, 4.7.25, 4.7.26, 4.7.29, 4.7.35 and 4.7.36. Reinstall the adapter using the same tools and retest the heater as above.

4.7.5 Combustion air inlet strength. Type II, MIL-H-62550/2 only. To determine conformance to 3.3.5, the heater shall be suspended with a suitable strap (less than 2.5 cm [1.0 inch] wide) from each of the inlet tubes listed below. Weights shall be attached so that the total suspended weight (heater plus weights) shall be at least 50 kg (110 lbs). Examine the heater for evidence of interfering parts, such as vibration, slow motor speeds, or sounds of rubbing parts. The heater shall be tested at the following combustion air inlet diameters:

a. 38.10 mm (1.50 in) diameter. Measure the torque necessary to rotate the combustion air inlet through the specified range on the applicable specification sheet. Suspend the heater as specified above. Remove the heater from the test setup. Inspect the heater and measure the torque necessary to rotate the combustion air inlet.

b. 43.18 mm (1.7 in) max diameter. Support heater from center (within 15 percent) of tube as specified.

4.7.6 Temperature.

4.7.6.1 High temperature. To determine conformance to storage requirements of 3.4.1.1 test the heater as specified by MIL-STD-202, method 108, condition B at not less than 71°C (160°F) in the position specified in table V. After storage reduce the heater temperature to 23 ± 10°C (73 ± 18°F) and test the heater as specified in 4.7.14, 4.7.18, 4.7.19, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.29, 4.7.35, 4.7.36 using DF-2 fuel. To determine conformance to combustion air requirements of 3.4.1.1 measure the smoke density as specified in ASTM D2156 after operating the heater for a minimum of one hour at high heat using combustion air at a minimum of 49°C (120°F).

4.7.6.2 Low temperature. To determine conformance to 3.4.1.2 test the heater in accordance with MIL-STD-810, method 502, procedure I. Preconditioned the heater by operating with DF-A

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fuel. Store the heater below -54°C (-65°F) for at least 12 hours in the position specified in table V. After storage, stabilize the heater at a temperature of -48 ± 3°C (-55 ± 5°F) and test the heater as specified in 4.7.14, 4.7.18, 4.7.19, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.29, 4.7.35, 4.7.36 using DF-A fuel. Repeat the test using DF-2 fuel at -4 ± 3°C (25 ± 5°F).

4.7.7 Humidity. To determine conformance to 3.4.2 test the heater in accordance with MIL-STD-202, method 106.

4.7.8 Salt fog. To determine conformance to 3.4.3 test the heater in accordance with MIL-STD-202, method 101, condition A.

4.7.9 Shock. To determine conformance to 3.4.4 test the heater, with no fuel applied, as specified by MIL-STD-202, method 213, condition A, except the pulse shall have an amplitude of 40 g's and a duration of 18 milliseconds. Mount the heater at the strap space locations shown on the applicable specification sheets.

4.7.10 Vibration. To determine conformance to 3.4.5 rigidly mount the heater at the strap space locations shown on the applicable specification sheet and subject it to vibration as specified by MIL-STD-202, method 204, condition A, except use the curve shown by figure 1. During testing the heater shall be started, purged, operated and monitored for intermittent operation. Conduct a resonant frequency search along each axis over a range of 5 to 500 hertz (Hz). Include 13-1/3 minutes dwell time at each resonance. If more than four resonances are present use the most severe. Cycle for 80 minutes on each axis including the resonance dwell time. Divide cycle time equally between operation at low and high heat.

4.7.11 Electromagnetic interference (EMI). To determine conformance to 3.4.6, the heater shall be subject to the tests specified by MIL-STD-462, notice 3. While it is exposed to the specified interference, it shall be operated as specified by 4.7.14 and 4.7.26. Starting and operating the heater may be simulated by eliminating fuel combustion. If this simulation is used, the contractor's test plan shall describe completely how he intends to simulate the heater's operation and demonstrate that his simulation is adequate to ensure representative heater operation during the specified testing.

4.7.12 Dust. To determine conformance to 3.4.7 a class II heater shall be operated at 28.0 ± 0.5 vdc at both heats for at least 100 hours at each heat. During operation fine dust shall be added to the heater's combustion air at the specified rate in accordance with MIL-STD-810, method 510. Repeat the above test

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except add coarse dust to the heater's fuel at a rate of not less than 1.5 grams (0.048 ounces) per hour, for a total of not less than 300 grams (9.6 ounces). Test the heater according to 4.7.25 upon completion of the dust testing.

4.7.13 Washdown. To determine conformance to 3.4.8, a class II heater shall be subjected to the following steps:

Step 1: Test setup. Horizontally mount the heater. Horizontally orient the combustion air outlet. Attach a stand pipe to the heater's combustion air outlet with adapter(s) as required. The stand pipe and the adapters shall be shaped to direct the test material (a suspension of test grit in water) to the heater's combustion air outlet. The stand pipe and adapter shall be at least 30 cm (12 inches) high with a volume of at least 4 liters (one gallon). The combustion air inlet may be sealed to prevent loss of test material.

Step 2: Test material. The test grit shall consist of 70 percent AC coarse (per SAE J726) and 30 percent 80 grit aluminum oxide. The test material shall be in suspension of at least 125 grams (4 ounces) test grit and 4 liters (one gallon) of water. The test grit shall be maintained in suspension as it is added to the set up's stand pipe.

Step 3: Procedure. Fill the stand pipe with test material to within one centimeter (0.5 inches) of the pipe's top (add water, as necessary, stirring to maintain the suspension). Simultaneously apply at least 275 kPa (40 psig) to all of the mixture's top surface and remove the seal from the combustion air inlet. Maintain pressure for at least three to five seconds. Wait for 30 minutes and attempt to start the heater. Monitor the control box run indicator for the specified flashing (see 3.5.2). Repeat the above starting attempt procedure at least three more times. Remove the heater from the test set up and drain it for less than 30 seconds. Subsequently, operate the heater with 28 $\pm$ 0.5 Vdc at high and low heat for at least two hours each.

Step 4: Reseal the combustion air inlet if necessary (Step 1) and repeat step 3 for at least four more times.

4.7.14 Starting cycle. To determine conformance to 3.5.1.1 place the control box switch in the "start" position for 2 $\pm$ 1 seconds. Return the switch to its "run" position. Monitor the control box run indicator. Repeat this test except keep the switch in the "start" position until the control box run indicator lights and then place the switch in the "run" position.

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4.7.15 Starting with fault. To determine conformance to 3.5.1.2, the heater shall be subjected to the tests a, b, c, and d below. These test shall be completed prior to durability and shock testing.

a. Move the control box's start-run switch to the start position and hold it there for at least 30 minutes. Start and operate the heater.

b. Remove the igniter from the heater and leave the heater's igniter circuit open. Move the control box's start-run switch to the start position and hold for 30 minutes. Measure and record the total fuel flow delivered to the heater.

c. Repeat the test from item b with the igniter circuit shorted out using a copper wire of at least 10 gauge and not longer than 30 cm (10 inches). Measure and record the total fuel flow to the heater.

d. Re-install the igniter and operate the heater.

4.7.16 Self diagnostics. To determine conformance to 3.5.2 monitor control box run indicator and induce system support and repair part failures. Examine heater for identification of fault.

4.7.17 Fuel compatibility. To determine conformance to 3.5.3.1 perform the durability test as specified in 4.7.39.

4.7.18 Fuel consumption. To determine conformance to 3.5.3.2 measure the fuel flow with 103 (+0, -7) kPa (15 [+0, -1] psig) fuel pressure applied to the fuel valve inlet. Repeat procedure with fuel pressure of 20.7 ± .7 kPa (3.0 ± 0.1 psig).

4.7.19 Fuel leakage. To determine conformance to 3.5.3.3 operate the heater for at least 5 minutes with a fuel pressure of 96.5 ± 6.7 kPa (14.0 ± 1 psig) applied to the fuel valve inlet. Subsequently remove the heater control box cover and examine the fuel circuit for fuel leakage.

4.7.20 Low fuel pressure. To determine conformance to 3.5.3.4, perform the following two steps:

Step 1. Before entering the start cycle, lower the fuel pressure at the fuel inlet to 27.6 ± 0.7 kPa (4.0 ± 0.1 psig). Start the heater and monitor the control box run indicator. Shut the heater off. Raise the fuel pressure to 55 ± 7 kPa (8 ± 1 psig). Start the heater and monitor the control box run indicator.

Step 2. Repeat Step 1 except operate the heater for one hour prior to start of test.

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4.7.21 Voltage. To determine conformance to 3.5.4 measure the applied voltage.

4.7.22 Transient voltage. To determine conformance to 3.5.4.1 subject the heater to spike and surge voltages described by MIL-STD-1275 as being generated during fault free and single fault operation of the vehicle electrical system. Apply the specified voltages at least 100 times during the off state of semiconductors in the heater which operate in the switching mode and are exposed to these voltages.

4.7.23 Current. To determine conformance to 3.5.5 measure the heater current draw.

4.7.24 Power loss. To determine conformance to 3.5.6, operate the heater for at least 95 minutes while monitoring the fuel flow. Remove the power from the heater control box. Measure the amount of fuel that flowed after power loss. Wait at least one hour. Restore power to the control box and move the start-off-run switch to the start position. Repeat this test 10 times.

4.7.25 Heat output. To determine conformance to 3.5.7 perform the following:

a) Assemble test setup as shown in figure 2.

b) Set restrictions and measure air flow per 4.7.26.

c) Measure input and output ventilation air temperature.

d) Run the heater until the ventilating outlet air temperature has stabilized.

e) When testing at ambient temperatures above 13°C (55°F) and the ventilating air outlet temperature is being limited by the heater's controls, the controls may be locked out if the ventilation outlet temperature is between 135 and 177°C (275 and 350°F).

f) Calculate the heat output using the following equation:

$H = V \times d \times Cp \times T$

H = Heat output in Watts (Btu/hr).
V = Ventilating air flow rate in cubic meters per second (m^3/sec) (cubic feet per hour [cu ft/hr]).
d = Density of ventilating air in kilograms per cubic meter (kg/m^3) (pounds per cubic foot [lb/cu ft]).

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Cp = Average specific heat of the ventilating air (from temperature rise across heater) in kilojoules per kilogram per degree C (kj/kg·°C) (Btu per lb per °F), including moisture content.

T = Ventilating air temperature rise across heater (outlet temperature, in degrees C(°F), minus inlet temperature, in degrees C(°F)).

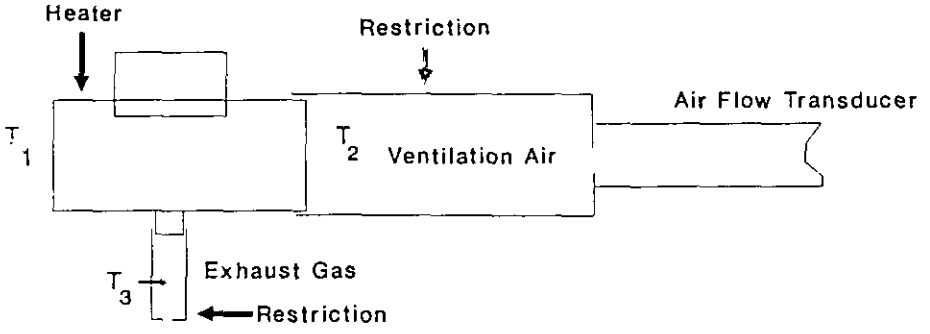


Figure 2. Heater set up for pressure loss testing. Seal transducer and restriction as required. Air flow measurement as specified.

T_1 = Ambient Temperature

T_2 = Ventilation Air Temperature. 42 ± 1 inch from heater and within 10% of of duct's center line.

T_3 = Exhaust Gas Temperature. 5 ± 1 inch from heater and within 10% of duct's center line.

4.7.26 Ventilating air discharge rate. To determine conformance to 3.5.8 perform the following:

a) Assemble heater test setup per figure 2.

b) Set restrictions as specified in table IX.

c) Measure the air flow and pressure flow losses in accordance with AMCA 210-74 or AMSE PTC 19.5.

d) During subsequent testing no changes shall be made in these restrictions.

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e) The contractor shall verify the coefficient of heat transfer of the ventilating air duct wall is less than 1.14 watts per square meter per degree C (0.20 Btu/hr per square foot per degree F)

f) Operate the heater and measure the air flow.

TABLE IX. Air circuit restrictions.

Class	Minimum Pressure Loss kPa (in H_2O)	Air Flow kl/min (scfm)
	Ventilation	
I	0.19 (0.75)	2.7 ± 0.14 (94 ± 5)
II	0.32 (1.30)	4.9 ± 0.14 (174 ± 5)
	Combustion	
I	0.19 (0.75)	0.235 ± 0.014 (8.3 ± .5)
II	0.24 (0.95)	0.410 ± 0.014 (14.5 ± .5)

4.7.27 Ventilation. Class II heaters only. To determine conformance to 3.5.8.1 measure air flow in the combustion and ventilation circuits in accordance with AMCA 210-74 or ASME PTC 19.5 with power applied to pin E of the heaters connector.

4.7.28 Seal leakage. Class II heaters only. To determine conformance to 3.5.8.2 assemble heater test set up as shown in figure 3. Install the heater's ventilation air outlet in a duct with a maximum diameter of 201 mm (7.94 inches) and operate the heater on high heat with the specified restriction of 4.7.26. Collect air leaking past the clearance between the heater and the duct in a plenum sealed between them. Measure the plenum's pressure and hold it within 0.012 kPa (0.05 in H_2O) of the heater's ventilation air inlet. Measure the air flow leaving the plenum in accordance with AMCA 210-74 or ASME PTC 19.5. Repeat the test with an air duct having a minimum diameter of 209mm (8.20 inches).

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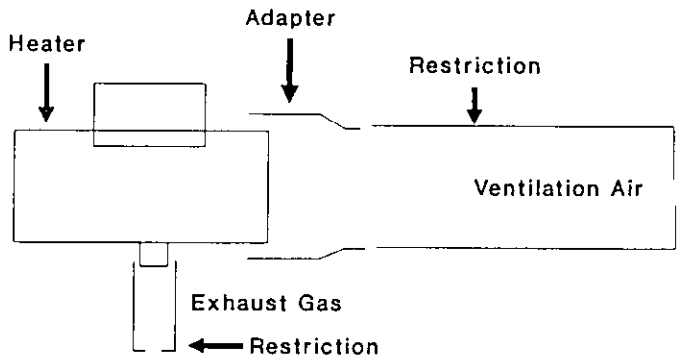


Figure 3. Test set up for seal leakage. Seal adapter to restriction as required. Adapter to heater dimension as specified and no sealing allowed.

4.7.29 Ventilating air restriction. To determine conformance to 3.5.8.3, measure the ventilating air flow in the duct, its temperature and heater fuel flow. Operate the heater at high heat into the duct used in 4.7.26, restricting the ventilating air flow until the heater limits the air's temperature between 150 and 177°C (300 and 350°F) or not more than 1.68 kl/min (60 scfm) (class I) or 3.36 kl/min (120 scfm) (class II). Continue to operate the heater for at least an additional 30 minutes. If applicable, within a period of not less than five minutes, continuously reduce the air flow to not more than 1.70 kl/min (60 scfm) (class I) or 3.40 kl/min (120 scfm) (class II). Air temperature shall remain within the specified limits. Continue to operate the heater for at least an additional 30 minutes. Then reduce the air flow to greater than 0.42 kl/min (15 scfm) (class I) or 0.85 kl/min (30 scfm) (class II). Note that the heater shut's off, stops fuel flow and enter its purge cycle without the ventilating air temperature exceeding 177°C (350°F).

4.7.30 Inlet over pressure. Class II heaters only. To determine conformance to 3.5.9 apply 2.0 kPa (8 in H_2O) pressure to the heater's combustion air inlet. Measure The combustion air flow in accordance with AMCA 210-74 or ASME PTC 19.5. Test the heater as specified in 4.7.25.

4.7.31 Combustion air flow. Class II heaters only. To determine conformance to 3.5.10.1 test the heater three times each as specified below.

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4.7.31.1 Reverse flow.

a. Apply at least 1.5 kPa (6.0 inches H_2O) to the heater's exhaust. Attempt to start the heater. Monitor the control box run indicator.

b. Operate the heater and apply 1.5 kPa (6.0 inches H_2O) to the heaters exhaust for 45 to 60 seconds. Remove the pressure from the exhaust for 3 to 4 minutes. Reapply the pressure to the exhaust for at least another 5 minutes. Monitor the fuel flow and the control box run indicator. Measure the combustion air inlet temperature within 3 cm (1.2 inches) of the inlet. Record the air inlet temperature and the time it takes the heater to enter the purge cycle.

4.7.31.2 Reduced flow.

a. Reduce air flow to less than 280 l/min (10 scfm) measured per ASTM 210-74 or ASTM PTC 19.5. Attempt to start the heater. Note if the heater starts and the control box run indicator condition.

b. Operate the heater for at least 1 hour. Slowly reduce the air flow to less than 280 l/min (10 scfm) measured per ASTM 210-74 or ASTM 9.5. During flow reduction, measure the smoke dot (per ASTM D2156) and monitor the control box run indicator. Record these conditions and the time to enter and duration the heater is in the purge cycle.

4.7.32 Combustion exhaust. To determine conformance to 3.5.10.2 measure the heater combustion exhaust temperature.

4.7.33 Combustion vapor exposure. For Type II heaters only. To determine conformance to 3.5.10.3 test the heater as specified below:

a. Heater configuration. Remove the heater adapters to convert to a Type I configuration. Measure the combustion air flow and maintain it to with $\pm$ 5% for entire test. Deliver a combustible mixture to the combustion air inlet and allow for observation of the luminous combustion from outside its body.

b. Chamber design and operation. The chamber shall be closed but for means of adding and removing the air, saturated vapor and draining spent test fluid. It shall be designed to keep the air from bubbling through the test fluid and to prevent fluid carry over. The test fluid shall wet the chamber's walls and shall not be recirculated or resprayed through the chamber. Sufficient fluid shall be sprayed into the chamber to form a steady stream

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of spent test fluid in the drain. While testing, saturation of the vapor shall be shown by warming the chamber's incoming air, cooling the chamber's exiting air within one degree °C (two degree °F) of the combustion air to condense a portion of the vapor and continuously remove the condensate.

c. Saturated vapor. The saturated vapor shall be generated by the chamber from lacquer thinner conforming to TT-T-266 and repeated with winter grade gasoline conforming to ASTM D439.

d. Combustion air. The heater's combustion air shall be mixed with a combustible gas (saturated vapor or propane, as applicable) to give a test mixture. An increasing amount of combustible gas shall be added to the combustion air linearly over a period of at least 20 minutes, starting at less than one percent combustible gas and ending with 100 percent saturated vapor or 25 percent propane.

4.7.34 Over temperature power shut off. To determine conformance to 3.5.11 restrict the ventilation air flow until the heater shuts off the fuel flow and enters the purge cycle (see 4.7.36). Monitor the voltage at pin C of the heater connector and record the value when the fuel flow is shut off.

4.7.35 Smoke density. To determine conformance to 3.5.12 measure the smoke density of the exhaust gasses in accordance with ASTM D2156 within seven minutes of ignition.

4.7.36 Purging cycle. To determine conformance to 3.5.13 stabilize the ventilating air temperature at high heat. Place the control box switch in the "off" position. Monitor and record the time for the exhaust temperature to drop to 177°C (350°F) and for the heater blower to automatically shut off.

4.7.37 Toxic fumes. To determine conformance to 3.5.14, prior to heater assembly, all heat exchangers shall be tested according to MIL-STD-1184, procedure 5, except in lieu of applying a vacuum to exterior of heat exchanger, apply pressure internally. As the last test of first article and control test, pressurize the exhaust tube with air at not less than 0.75 kPa (3 in H_2O) pressure, block the purpose openings, and measure the air flow rate in accordance with AMCA 210-74 or ASME PTC 19.5.

4.7.38 Reliability and durability verification. To verify the heater assembly meets the reliability and durability requirements of 3.6, three (3) heater assemblies shall be placed on test for 2500 hours each. There shall be no more than six (6) chargeable reliability failures (cumulative) and zero (0) chargeable durability failures on the three (3) samples.

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4.7.39 Reliability and durability testing. To determine conformance to 3.6, operate the heater, as specified by paragraphs a through g, for the four phases listed below.

a. Monitor self-diagnostics throughout.

b. Accumulate total operation hours of only cycle "on" time specified in table X. Shut the heater off for at least ten minutes after each "on" cycle.

c. Maintain fuel at the ambient temperature specified in each phase.

d. Operate the heater with restrictions in 4.7.25 and 4.7.26.

e. Install a filter conforming to MIL-F-45356, type I or II, class I in the fuel line leading to the heater. Verify fuel pressure of 8 ± 1 psig at filter inlet. Add fine contaminate per MIL-F-45356 to the fuel up stream of the filter at a rate less than 0.5 grams per hour. Add 100 grams of contaminate in each of the first three phases (300 g total). when the control box run indicator flashes and the heater enters the self diagnostic mode, replace the filter and continue adding contaminate.

(Note: Replacement of this clogged filter does not constitute a heater failure.)

f. Verify tests 4.7.14, 4.7.21, 4.7.23 and 4.7.36 during start and operation of each cycle. Drop the applied voltage (4.7.21) to 19 ± 0.5 Vdc for 2.5 ± 0.5 seconds for every 25th start cycle in Phases I and II.

g. Verify tests 4.7.18, 4.7.19, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.29, 4.7.32 and 4.7.35 after each phase. Do not add contaminant.

PHASE I - Using DF-2 fuel at ambient temperature of 23 ± 10°C (73 ± 18°F)

Step a: Set heater at low heat and operate for 25 cycles with an applied voltage of 23 ± 0.5 Vdc.
Step b: Repeat step a using an applied voltage of 28.0 ± 0.5 Vdc.
Step c: Set heater at high heat and repeat steps a and b.
Step d: Repeat steps a through c six times (600 total cycles).

PHASE II - Using DF-1 fuel at ambient temperature of -32.0 ± 3°C (-25 ± 5°F) repeat steps a through d three times (300 total cycles), and substitute 30.0 ± 0.5 Vdc for 28.0 ± 0.5 Vdc.

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PHASE III- Using DF-A fuel at ambient temperature of -48.0 ± 3°C (-55 ± 5°F) repeat steps a through d three times (300 total cycles), substitute less than 19 ± 0.5 Vdc in steps a, and 30 ± 0.5 Vdc in step b.

PHASE IV - Repeat Phases I through III a second time (2400 total cycles).

TABLE X. Durability test times for class I and II heaters.

Heater Class	Phase	Cycles (total)	Cycles on time (minutes)	Operation hours total
I	I	1200	30	600
I	II	600	30	300
I	III	600	30	300
II	I	1200	95	1900
II	II	600	30	300
II	III	600	30	300

4.7.39.1 Reduced durability. For control test only. The test shall consist of 100 cycles each of phases I, II and III (see 4.7.39). Addition of dust not required.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Application information. The heater furnished under this specification is intended to supply warm air to the crew compartment of military vehicles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of specification sheet (see 3.2). Heaters conforming to Type I, Class II or MIL-H-62550/1 should not be ordered. In their lieu, heaters conforming

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to Type II, Class II or MIL-H-62550/2 should be ordered. The latter heaters are field convertible (see 3.3.3) to the former.

c. If first article is required (see 3.1).
d. If responsibility for inspection and the place of inspection is other than the contractors (see 4.1).
e. Arrangements for first article inspection, approval of test results, and disposition of first article (see 4.3).
f. Government's rights and conditions for waiving first article inspection (see 6.3).
g. Selection of applicable level and packaging requirements (see 5.1).
h. Sampling table, if other than as specified (see 4.4.1.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the first article samples should be first article sample, preproduction sample, initial production sample, first production item, or standard production item from the contractor's current inventory. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Supersession data.

Superseded Part	New Part (MIL-H-62550)
M46792/2	Type I, Class I
7748716	Type I, Class II
11601809	Type I, Class II
11669489	Type II, Class II
11669489-1	Type II, Class II
11669490	Type II, Class II
11669490-1	Type II, Class II
11669496	Type I, Class II

6.5 Criticality. This is not a critical item.

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6.6 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Military Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.7 Process average. Sampling may be initiated if the process average value for the first 100 heaters inspected is less than 1% for major defects and less than 4% for minor defects. The process average should be calculated as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of heaters inspected}} \times 100\%$$

6.8 Subject term (key word) listing.

diagnostics, self
air flow
ventilation
over pressure
smoke density
combustion vapor

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 2540-A037)

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